

If we let A_1 = area under the graph of f from 0 to $\frac{1}{2}$
 and A_2 = area under the graph of f from 0 to 2,
 the area A can be found by subtracting A_1 from A_2 :

$$A = A_2 - A_1$$

In Example 7 we found that the area under the graph of $y = x^3$ from 0 to b is $\frac{1}{4}b^4$. Hence, using $b = \frac{1}{2}$ for A_1 and $b = 2$ for A_2 yields

$$A = \frac{2^4}{4} - \frac{(\frac{1}{2})^4}{4} = 4 - \frac{1}{64} \approx 3.98.$$

EXERCISES 5.3

Exer. 1–8: Evaluate the sum.

$$1 \sum_{j=1}^4 (j^2 + 1)$$

$$2 \sum_{j=1}^4 (2^j + 1)$$

$$3 \sum_{k=0}^5 k(k-1)$$

$$4 \sum_{k=0}^4 (k-2)(k-3)$$

$$5 \sum_{n=1}^{10} [1 + (-1)^n]$$

$$6 \sum_{n=1}^4 (-1)^n \left(\frac{1}{n}\right)$$

$$7 \sum_{i=1}^{50} 10$$

$$8 \sum_{k=1}^{1000} 2$$

Exer. 9–12: Express the sum in terms of n (see Example 4).

$$9 \sum_{k=1}^n (k^2 + 3k + 5)$$

$$10 \sum_{k=1}^n (3k^2 - 2k + 1)$$

$$11 \sum_{k=1}^n (k^3 + 2k^2 - k + 4)$$

$$12 \sum_{k=1}^n (3k^3 + k)$$

Exer. 13–18: Express in summation notation.

$$13 \ 1 + 5 + 9 + 13 + 17$$

$$14 \ 2 + 5 + 8 + 11 + 14$$

$$15 \ \frac{1}{2} + \frac{2}{5} + \frac{3}{8} + \frac{4}{11}$$

$$16 \ \frac{1}{4} + \frac{2}{9} + \frac{3}{14} + \frac{4}{19}$$

$$17 \ 1 - \frac{x^2}{2} + \frac{x^4}{4} - \frac{x^6}{6} + \cdots + (-1)^n \frac{x^{2n}}{2n}$$

$$18 \ 1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \cdots + \frac{x^n}{n}$$

Exer. 19–22: Let A be the area under the graph of the given function f from a to b . Approximate A by dividing $[a, b]$ into subintervals of equal length Δx and using (a) A_{IP} and (b) A_{CP} .

$$19 \ f(x) = 3 - x; \quad a = -2, \quad b = 2; \quad \Delta x = 1$$

$$20 \ f(x) = x + 2; \quad a = -1, \quad b = 4; \quad \Delta x = 1$$

$$21 \ f(x) = x^2 + 1; \quad a = 1, \quad b = 3; \quad \Delta x = \frac{1}{2}$$

$$22 \ f(x) = 4 - x^2; \quad a = 0, \quad b = 2; \quad \Delta x = \frac{1}{2}$$

$$\text{[C]} \ 23 \ f(x) = \sqrt{\sin x}; \quad a = 0, \quad b = 1.5; \quad \Delta x = 0.15$$

$$\text{[C]} \ 24 \ f(x) = \frac{1}{\sqrt{x^3 + 1}}; \quad a = 0, \quad b = 3; \quad \Delta x = 0.3$$

Exer. 25–30: Refer to Examples 6 and 7. Find the area under the graph of the given function f from 0 to b using (a) inscribed rectangles and (b) circumscribed rectangles.

$$25 \ f(x) = 2x + 3; \quad b = 4$$

$$26 \ f(x) = 8 - 3x; \quad b = 2$$

$$27 \ f(x) = 9 - x^2; \quad b = 3$$

$$28 \ f(x) = x^2; \quad b = 5$$

$$29 \ f(x) = x^3 + 1; \quad b = 2$$

$$30 \ f(x) = 4x + x^3; \quad b = 2$$

Exer. 31–32: Refer to Example 7. Find the area under the graph of f corresponding to the interval (a) $[1, 3]$ and (b) $[a, b]$.

$$31 \ f(x) = x^3$$

$$32 \ f(x) = x^3 + 2$$

5.4 THE DEFINITE INTEGRAL

In Section 5.3 we defined the area under the graph of a function f from a to b as a limit of the form

$$\lim_{\Delta x \rightarrow 0} \sum_{k=1}^n f(w_k) \Delta x.$$